GROUPS OF CLASS TRANSPOSITIONS WITH PRESCRIBED PRIME DIVISORS OF THE MODULI

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ABSTRACT. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be sets of odd primes. Problem 21.75 asks whether the inclusion $\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\rangle \lneq \mathrm{CT}_{\mathcal{P}_1\cup\mathcal{P}_2}(\mathbb{Z})$ is strict whenever neither of the sets $\mathcal{P}_1, \mathcal{P}_2$ is contained in the other. We prove that, for arbitrary $\mathcal{P}_1$ and $\mathcal{P}_2$, one has $\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\rangle = \mathrm{CT}_{\mathcal{P}_1\cup\mathcal{P}_2}(\mathbb{Z})$.

1. INTRODUCTION

For $m \geqslant 1$ and $0 \leqslant a < m$, let $a(m) = a + m\mathbb{Z}$ denote the residue class of a modulo m. Two classes $a(m)$ and $b(n)$ are disjoint if and only if $a \not\equiv b \pmod{\gcd(m,n)}$.

Let $a(m)$ and $b(n)$ be disjoint. The class transposition associated with them is the permutation

$$\tau_{a(m),b(n)} = \prod_{k\in\mathbb{Z}} (a+mk, b+nk) \in \mathrm{Sym}(\mathbb{Z}).$$

If $\mathcal{P}$ is a set of odd primes, then $\mathrm{CT}_{\mathcal{P}}(\mathbb{Z})$ denotes the group generated by all class transpositions $\tau_{a(m),b(n)}$ for which every odd prime divisor of mn belongs to $\mathcal{P}$. In particular, powers of two are allowed in the moduli for every $\mathcal{P}$.

The following question appears as Problem 21.75 in the Kourovka Notebook [2].

Question 1.1. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be distinct sets of odd primes, neither of which is contained in the other. Is it true that

$$\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\rangle \lneq \mathrm{CT}_{\mathcal{P}_1\cup\mathcal{P}_2}(\mathbb{Z})?$$

It is clear that $\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\rangle \leqslant \mathrm{CT}_{\mathcal{P}_1\cup\mathcal{P}_2}(\mathbb{Z})$. In this paper we prove the reverse inclusion, thereby giving a negative answer to Problem 21.75.

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2. Class Transpositions Separated Modulo a Power of Two

Fix a power of two $R = 2^k \geqslant 4$. A class transposition $\tau_{a(m),b(n)}$ will be called R-separated if

$$R \mid m, \qquad R \mid n, \qquad a \not\equiv b \pmod{R}.$$

The last condition, in particular, ensures that the classes $a(m)$ and $b(n)$ are disjoint.

Lemma 2.1. Let the class transposition $\tau_{a(m),b(n)}$ be R-separated. Suppose that q is an odd prime and $q \mid n$. Then there exists a class transposition $\tau_{b_0(Rq),c(R)}$ such that

$$\tau_{b_0(Rq),c(R)}\tau_{a(m),b(n)}\tau_{b_0(Rq),c(R)} = \tau_{a(m),b'(n/q)}. \tag{2.1}$$

The class transposition on the right-hand side is again R-separated.

Proof. Since $R \mid n$, $q \mid n$, and q is odd, the number $d = Rq$ divides n. Divide b by d with remainder:

$$b = b_0 + dt, \qquad 0 \leqslant b_0 < d.$$

The inequality $0 \leqslant b < n$ implies that

$$0 \leqslant t < \frac{n}{d}. \tag{2.2}$$

Since $a \not\equiv b \pmod{R}$ and $R \geqslant 4$, we may choose $c \in \{0, \ldots, R-1\}$ such that

$$c \not\equiv a \pmod{R}, \qquad c \not\equiv b \pmod{R}. \tag{2.3}$$

Because $b_0 \equiv b \pmod{R}$, the condition $a \not\equiv b \pmod{R}$ together with (2.3) shows that the classes

$$a(m), \quad b_0(Rq), \quad c(R)$$

are pairwise disjoint. Thus the class transposition $\tau_{b_0(Rq),c(R)}$ is well defined. It fixes the class $a(m)$ pointwise.

Let us compute the image of the class $b(n)$ under $\tau_{b_0(Rq),c(R)}$. For every $k \in \mathbb{Z}$, we have

$$\begin{aligned} b + nk &= b_0 + Rq\left(t + \frac{n}{Rq}k\right) \\ &\overset{\tau_{b_0(Rq),c(R)}}{\longmapsto} c + R\left(t + \frac{n}{Rq}k\right) \\ &= (c + Rt) + \frac{n}{q}k. \end{aligned} \tag{2.4}$$

Set $b' = c + Rt$. By (2.2) and $0 \leqslant c < R$, we obtain

$$0 \leqslant b' < R + R\left(\frac{n}{Rq} - 1\right) = \frac{n}{q}.$$

Since $\tau_{b_0(Rq),c(R)}$ fixes $a(m)$, it follows that

$$\tau_{b_0(Rq),c(R)}\tau_{a(m),b(n)}\tau_{b_0(Rq),c(R)} = \tau_{a(m),b'(n/q)}.$$

Finally,

$$b' = c + Rt \equiv c \not\equiv a \pmod{R},$$

so the class transposition on the right remains R-separated. □

Lemma 2.2. Suppose that every odd prime divisor of mn belongs to $\mathcal{P}_1 \cup \mathcal{P}_2$. Then every R-separated class transposition $\tau_{a(m),b(n)}$ belongs to the group $\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \rangle$.

Proof. Suppose that $q \in \mathcal{P}_2 \setminus \mathcal{P}_1$ divides one of the moduli. Applying Lemma 2.1 (with the two classes interchanged if necessary), we conjugate by a class transposition whose moduli are Rq and R, thereby decreasing the exponent of q in that modulus by one. This conjugating class transposition belongs to $\mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})$, since q is the only possible odd prime divisor of its moduli.

Continue this process. After each step, both moduli are still divisible by R, since q is odd, and the representatives of the two classes remain distinct modulo R. After finitely many steps, we remove from both moduli all prime divisors belonging to $\mathcal{P}_2 \setminus \mathcal{P}_1$. The result is an R-separated class transposition τ_0 such that every odd prime divisor of its moduli belongs to $\mathcal{P}_1$. Hence

$$\tau_0 \in \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}) \leqslant \langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \rangle.$$

The original class transposition is conjugate to τ_0 by a product of elements of $\mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \leqslant \langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \rangle$. Therefore it also belongs to $\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \rangle$. □

3. Horizontal Class Transpositions

A class transposition $\tau_{a(m),b(n)}$ is called horizontal if $m = n$.

Lemma 3.1. Suppose that $R \mid M$ and that every odd prime divisor of M belongs to $\mathcal{P}_1 \cup \mathcal{P}_2$. Then, for any distinct $x, y \in \{0, \dots, M-1\}$, one has

$$\tau_{x(M),y(M)} \in \langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \rangle.$$

Proof. If $x \not\equiv y \pmod{R}$, apply Lemma 2.2.

Now suppose that $x \equiv y \pmod{R}$. Choose $z \in \{0, \dots, R-1\}$ such that $z \not\equiv x \pmod{R}$. Since $R \mid M$, we have $0 \leqslant z < M$. The classes $x(M), y(M), z(M)$ are pairwise distinct and hence pairwise disjoint. Therefore

$$\tau_{x(M),y(M)} = \tau_{x(M),z(M)} \tau_{y(M),z(M)} \tau_{x(M),z(M)}. \tag{3.1}$$

Both class transpositions on the right-hand side are R-separated and hence belong to $\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \rangle$ by Lemma 2.2. □

4. A Matching Lemma

A matching in a graph is a set of pairwise nonincident edges. It covers a vertex if that vertex is incident with one of its edges, and it is perfect if it covers every vertex of the graph. For a bipartite graph with parts X and Y and a subset $S \subseteq X$, let $N(S) \subseteq Y$ denote the set of all vertices adjacent to at least one vertex of S. We recall Hall's classical marriage theorem.

Theorem 4.1 (Hall)**.** *Let Γ be a finite bipartite graph with parts X and Y. There exists a matching that covers every vertex in X if and only if*

$$|N(S)| \geqslant |S|$$

for every subset $S \subseteq X$. In particular, if $|X| = |Y|$ and this condition holds, then Γ has a perfect matching.

Lemma 4.2. Let $a, b \in \mathbb{Z}$, $m, n \in \mathbb{N}$, and $R \in \mathbb{N}$. If

$$\gcd(m, R) + \gcd(n, R) \leqslant R, \tag{4.1}$$

then there exists a permutation π of the set $\{0, \dots, R-1\}$ such that

$$a + mj \not\equiv b + n\pi(j) \pmod{R} \qquad (0 \leqslant j < R). \tag{4.2}$$

Proof. For every $c \in \mathbb{Z}/R\mathbb{Z}$, set

$$\alpha_c = \big|\{j \in \{0, \dots, R-1\} : a + mj \equiv c \pmod{R}\}\big|,$$
$$\beta_c = \big|\{j \in \{0, \dots, R-1\} : b + nj \equiv c \pmod{R}\}\big|.$$

In particular,

$$\alpha_c \leqslant \gcd(m, R), \qquad \beta_c \leqslant \gcd(n, R).$$

By (4.1), we obtain

$$\alpha_c + \beta_c \leqslant R \qquad (c \in \mathbb{Z}/R\mathbb{Z}). \tag{4.3}$$

Construct a bipartite graph whose left and right parts are two copies of $\{0, \dots, R-1\}$. Join a left vertex j to a right vertex k if and only if

$$a + mj \not\equiv b + nk \pmod{R}.$$

We verify Hall's condition.

Let S be a set of left vertices. Hall's condition is immediate for $S = \varnothing$, so assume that $S \neq \varnothing$. If the values $a + mj \pmod{R}$, $j \in S$, include at least two distinct residues, then every right vertex is adjacent to at least one vertex in S. Indeed, the residue corresponding to a right vertex cannot coincide with two distinct residues simultaneously. Hence, in this case, $N(S)$ is the entire right part and

$$|N(S)| = R \geqslant |S|.$$

If, on the other hand, all vertices in S have the same residue c, then $|S| \leqslant \alpha_c$, while the neighbors of S are precisely the right vertices whose residues differ from c. Consequently, using (4.3), we obtain

$$|N(S)| = R - \beta_c \geqslant \alpha_c \geqslant |S|.$$

Thus Hall's condition holds for every S. The graph therefore has a perfect matching. The corresponding bijection $j \mapsto \pi(j)$ satisfies (4.2). $\square$

5. Proof of the Main Result

The following proposition was proved in [1].

Proposition 5.1. Let $\tau_{a(m),b(n)}$ be a class transposition. Then:

(1) the permutation $\tau_{(a+mj)(Km),(b+nj)(Kn)}$ is a class transposition for all integers j and K such that $0 \leqslant j \leqslant K-1$;
(2) for every natural number K, one has

$$\tau_{a(m),b(n)} = \prod_{j=0}^{K-1} \tau_{(a+mj)(Km),(b+nj)(Kn)}.$$

Theorem 5.2. *For any sets $\mathcal{P}_1$ and $\mathcal{P}_2$ of odd primes, one has*

$$\left\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \right\rangle = \mathrm{CT}_{\mathcal{P}_1 \cup \mathcal{P}_2}(\mathbb{Z}).$$

Proof. Consider an arbitrary generator

$$T = \tau_{a(m),b(n)} \in \mathrm{CT}_{\mathcal{P}_1 \cup \mathcal{P}_2}(\mathbb{Z}). \tag{5.1}$$

Choose k sufficiently large that $R = 2^k$ satisfies

$$R \geqslant 4, \qquad \gcd(m, R) + \gcd(n, R) \leqslant R. \tag{5.2}$$

By Proposition 5.1,

$$T = \prod_{j=0}^{R-1} \tau_{(a+mj)(Rm),(b+nj)(Rn)}. \tag{5.3}$$

Set

$$A_j = (a + mj)(Rm), \qquad B_j = (b + nj)(Rn).$$

All factors in (5.3) have pairwise disjoint supports.

By Lemma 4.2, there exists a permutation $\pi \in S_R$ such that

$$a + mj \not\equiv b + n\pi(j) \pmod{R} \qquad (0 \leqslant j < R). \tag{5.4}$$

Consider the permutation

$$U = \prod_{j=0}^{R-1} \tau_{A_j, B_{\pi(j)}}. \tag{5.5}$$

By (5.4), every class transposition in this product is R-separated. Its moduli are Rm and Rn, and their odd prime divisors belong to $\mathcal{P}_1 \cup \mathcal{P}_2$. Therefore, by Lemma 2.2, all factors in (5.5) belong to $\left\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \right\rangle$. Hence $U \in \left\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z}) \right\rangle$.

We adopt the convention that, in a product of permutations, the rightmost factor acts first. Set $V = TU$. We now describe the action of V. If

$$x = a + mj + Rm\ell \in A_j,$$

then

$$x \overset{U}{\longmapsto} b + n\pi(j) + Rn\ell \overset{T}{\longmapsto} a + m\pi(j) + Rm\ell.$$

Consequently,

$$V\colon A_j \longrightarrow A_{\pi(j)}$$

with the parameter ℓ unchanged. Similarly, if $x \in B_j$, then

$$V\colon B_j \longrightarrow B_{\pi^{-1}(j)}$$

with the parameter unchanged. Outside $a(m) \cup b(n)$, both T and U act identically.

Write the permutation $\pi \in S_R$ as a product of transpositions:

$$\pi = (i_1\, j_1)\cdots(i_t\, j_t).$$

The action of V described above therefore gives

$$V = \big(\tau_{A_{i_1},A_{j_1}} \cdot \ldots \cdot \tau_{A_{i_t},A_{j_t}}\big)\big(\tau_{B_{i_t},B_{j_t}} \cdot \ldots \cdot \tau_{B_{i_1},B_{j_1}}\big). \tag{5.6}$$

Indeed, the first factor induces the permutation π on the classes $A_0, \ldots, A_{R-1}$ and fixes all classes B_j, whereas the second induces the permutation π^{-1} on the classes $B_0, \ldots, B_{R-1}$ and fixes all classes A_j.

Every transposition $\tau_{A_{i_\nu},A_{j_\nu}}$ is a horizontal class transposition of modulus Rm, and every $\tau_{B_{i_\nu},B_{j_\nu}}$ is a horizontal class transposition of modulus Rn. Both moduli are divisible by R, and all their odd prime divisors belong to $\mathcal{P}_1 \cup \mathcal{P}_2$. Therefore, by Lemma 3.1, all factors in (5.6) belong to $\big\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\big\rangle$. Hence $V \in \big\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\big\rangle$.

Thus every generator of $\mathrm{CT}_{\mathcal{P}_1\cup\mathcal{P}_2}(\mathbb{Z})$ belongs to $\big\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\big\rangle$. Together with the evident reverse inclusion, this yields

$$\big\langle \mathrm{CT}_{\mathcal{P}_1}(\mathbb{Z}), \mathrm{CT}_{\mathcal{P}_2}(\mathbb{Z})\big\rangle = \mathrm{CT}_{\mathcal{P}_1\cup\mathcal{P}_2}(\mathbb{Z}),$$

as required. □

References

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[2] E. I. Khukhro and V. D. Mazurov (eds.), *The Kourovka Notebook: Unsolved Problems in Group Theory*.

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